

Affective Reconsolidation & Plasticity Engine (ARPE): A Neurocognitive Model of Emotion-Driven Memory Updating and Structural Plasticity

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August 2026

Abstract

The Affective Reconsolidation and Plasticity Engine (ARPE) proposes an exploratory neurocognitive model for long-term memory updating and structural change. It models how emotionally salient mismatch may transiently destabilize existing cognitive-emotional structures, allowing them to be retagged and rewritten before restabilization. ARPE formalizes this process as a threshold-governed update sequence in which affective charge and resonance error jointly determine whether a structure remains stable, enters a labile window, updates partially, or restabilizes in modified form. The framework is intended to describe memory traces, beliefs, intentions, schemas, and personal narratives as stabilized structures that can become revisable under bounded conditions. It generates testable predictions for memory-reactivation and emotion-induction paradigms, physiological and neural reconsolidation markers, longitudinal coherence measures, and delayed behavior-transfer tests. Rather than claiming that emotional intensity alone drives adaptive change, ARPE positions structured affective error as a candidate mechanism for durable psychological updating.

As an exploratory framework, ARPE describes how stabilized psychological structures can remain both durable and revisable. Its central distinction is between activation and structural change: a structure may be recalled, challenged, discussed, or emotionally stirred without being modified. ARPE proposes that lasting change requires a threshold-governed sequence in which reactivation, affective salience, resonance error, integration, and coherence restoration occur together.

1.0 Introduction

How do memories, beliefs, intentions, and personal narratives change over time? Existing accounts of memory reconsolidation describe windows of instability, but they often leave the affective and predictive conditions of durable updating under-specified. The Affective Reconsolidation and Plasticity Engine (ARPE) addresses this gap by proposing an exploratory neurocognitive model in which reconsolidation windows become most relevant when emotionally salient prediction error makes a stabilized structure temporarily revisable.

During these windows, existing cognitive-emotional structures become labile, receive new affective tagging, and may be rewritten with updated parameters. The result is a structured account of plasticity: old patterns may weaken, new patterns may form, and the system can change while preserving enough coherence to remain continuous. ARPE reframes psychological change as an active, emotion-gated updating process rather than as passive consolidation or the simple accumulation of new information.

ARPE's contribution is not that emotion, memory, or prediction error individually influence change. Its contribution is the proposed interaction among them: a stabilized structure becomes revisable when reactivation, affective salience, and mismatch converge inside a bounded update window. This gives the framework a specific explanatory target: durable structural change that cannot be explained by exposure, persuasion, mood shift, or repetition alone.

In this view, memory, belief, intention, schema, and narrative are treated as stabilized cognitive-emotional structures that may share a common update logic despite differing in content, scope, and timescale.

ARPE defines a shared update logic that may apply across therapeutic change, learning, trauma recovery, identity revision, and narrative restructuring without reducing itself to any one of those domains. It is positioned at the mechanism level: it asks when a stabilized structure becomes available for change, which conditions determine whether update occurs, how new information is affectively bound, and how the system prevents destabilization from spreading into incoherence.

This position makes ARPE exploratory in two senses. First, it proposes a mechanism that can be tested across multiple domains rather than limiting itself to one intervention model. Second, it deliberately leaves implementation details open so that different laboratories, clinical contexts, and computational systems can instantiate the structure using domain-appropriate measures. A study of autobiographical memory may operationalize the structure differently from a study of skill learning or identity change, but each must preserve the same core relation: a stabilized structure becomes revisable only when affective error opens a bounded window and new input is bound into that window.

The manuscript proceeds in six parts. Section 2 defines the core ARPE mechanism and its pseudo-formal update relation. Section 3 describes the processing sequence from destabilization through restabilization. Section 4 translates the mechanism into testable predictions and discriminator logic. Section 5 defines limitations, competing explanations, and staged validation priorities. Section 6 synthesizes ARPE as an independent plasticity framework for emotion-driven structural updating. The appendices preserve symbols, constraints, update regimes, measurement scaffolding, and exploratory plasticity profiles.

2.0 Core Mechanism

ARPE operates through bounded reconsolidation windows that open only when a stabilized structure is reactivated under sufficient affective charge and resonance error. In this framework, resonance error refers to a meaningful mismatch between the current structure and incoming input: the old structure no longer fits the evidence, context, outcome, or interpretation now being encountered. Within such a window, five kinds of information remain conceptually distinct: the standing structure, new input, affective charge, resonance error, and the coherence of the structure itself.

Outside these windows the structure is stable, and reactivation on its own does not alter it. What distinguishes ARPE is therefore not plasticity as such, but the condition under which a stabilized structure becomes available for rewriting and the sequence through which it is returned to a stable form. Sections 2.3 through 2.6 describe that sequence; Section 3.0 details how it unfolds across a window.

ARPE distinguishes ordinary activation from destabilizing activation. Ordinary activation occurs whenever a structure is recalled, rehearsed, triggered, or applied without becoming available for modification. Destabilizing activation occurs when the structure is re-entered under sufficient affective charge and mismatch. The mismatch marks the structure as inaccurate, incomplete, or no longer adequate; the affective charge marks the mismatch as important enough to reorganize the system around. Only when these forces converge does the model predict a bounded plasticity interval.

2.1 Mechanism Scope and Structural Unit

The structural unit of ARPE is any stabilized cognitive-emotional configuration that can be reactivated, destabilized, retagged, and restabilized. This includes individual memories, belief clusters, self-evaluations, recurring intentions, avoidance patterns, role expectations, and narrative interpretations. The model does not require these structures to share one biological format or storage location. It requires only that they function as stabilized units: they can be re-entered, they resist casual change, they guide future interpretation, and they can be modified under bounded plasticity conditions.

This definition prevents ARPE from becoming restricted to a narrow memory-only model while also preventing overextension. A temporary mood, passing thought, or momentary impulse is not automatically an ARPE structure unless it has enough stability to recur and influence later processing. Conversely, a structure does not need to be permanent or consciously endorsed to qualify. An avoidance response, a body-level threat expectation, or a repeated narrative assumption may function as a structure even if the person cannot fully articulate it.

The scope of the mechanism is therefore bounded by traceability. If a claimed change cannot be traced to a destabilized target structure, ARPE should not be invoked as the primary explanation. If the target structure can be identified before the event, shown to become labile during the event, and shown to behave differently after restabilization, ARPE becomes a stronger interpretive candidate. This traceability requirement keeps the framework falsifiable and prevents it from absorbing all forms of learning or emotional change.

2.2 Core Components and Update Structure

The update relation is stated in pseudo-formal terms. It names the components involved and the threshold condition under which updating is permitted; it does not assign numerical values, define a runnable algorithm, or claim that the relation is a complete computational model.

$$S'(t+1) = \text{Reconsolidate}[S(t), I(t), A(t), E(t), C(t)] \text{ if } A(t) \times E(t) \geq \theta R$$

$$S'(t+1) = S(t) \text{ if } A(t) \times E(t) < \theta R$$

Where Reconsolidate refers to the destabilization, tagging, rewriting, and restabilization sequence described in Section 3.0. The first expression states the model condition under which rewriting is expected: affective charge and resonance error jointly meet or exceed the reconsolidation threshold. The second expression states that, when this condition is not met, the structure remains stable rather than being rewritten.

Parameter values, estimation procedures, measurement channels, and runnable defaults are not specified here. The present manuscript is concerned only with the structure of the update relation; any institutional edition may render that logic in operational form.

2.3 Reconsolidation Window Opening

A reconsolidation window opens when a stabilized structure is reactivated in the presence of sufficient affective charge and resonance error. Reactivation alone is not enough. A memory can be recalled many times without being rewritten, a belief can be challenged many times without changing, and an intention can be reviewed many times without becoming stable. ARPE proposes that the system must encounter a discrepancy that matters affectively. The discrepancy creates update pressure, while the emotional charge gives that pressure reorganizing force.

The window is bounded because the system cannot remain indefinitely labile without risking loss of coherence. ARPE therefore treats destabilization as a temporary state. During this interval the structure becomes flexible enough to accept new input, but not so dissolved that it loses identity. This distinction is central: the model does not describe random emotional volatility. It describes controlled structural availability, where the system permits revision only under conditions that preserve continuity between the old structure and the updated one.

In exploratory terms, window opening can be understood as a threshold transition from stability to update-readiness. Below threshold, the structure remains intact and may continue to be rehearsed without rewriting. At threshold, the structure becomes revisable. Above threshold, update may proceed if tagging and integration occur. Far above threshold, however, the system may enter overload, producing defensive closure or compensatory stabilization rather than adaptive rewriting. ARPE therefore predicts an optimal update zone rather than a simple rule that more emotion always produces more change.

2.4 Affective Tagging & Rewriting

Affective tagging is the process by which new input becomes bound to the destabilized structure under an updated emotional marker. The new input may be corrective evidence, a new interpretation, a relational experience, successful action, failed expectation, or internally generated reframing. What matters is that the input is not merely noticed; it must be attached to the labile structure while the window is open. Without this binding, the system may experience emotion and contradiction without producing durable reorganization.

Rewriting occurs when the tagged input alters the structure's internal weighting. A fear memory may lose threat weighting, a belief may lose certainty, an intention may gain motivational strength, or a narrative may acquire a new explanatory frame. ARPE does not require that the original structure disappear. More often, the structure is transformed: its emotional charge, predictive expectations, retrieval priority, or identity relevance shift enough that future activation produces a different system state.

This distinguishes ARPE from simple exposure, persuasion, repetition, or insight. Exposure without affective retagging may habituate response but leave structure intact. Persuasion without destabilization may add information without rewriting the prior. Repetition without mismatch may consolidate the existing structure. Insight without integration may produce temporary clarity that fails to alter later behavior. ARPE predicts its strongest form of durable structure-specific change when reactivation, affective charge, mismatch, tagging, and restabilization occur as a coherent sequence.

2.5 Restabilization

Restabilization is the process by which the updated structure closes its labile interval and returns to coherent standing form. The system must preserve enough continuity to recognize the structure as the same general memory, belief, intention, schema, or narrative, while incorporating enough change that future activation produces a different affective or interpretive outcome. Restabilization therefore marks the transition from temporary plasticity to durable update.

The updated structure can propagate change through related structures because cognitive-emotional systems are not isolated. A revised memory may alter a belief; a softened belief may change an intention; a new intention may reorganize self-narrative; a narrative shift may change which memories are retrieved in future contexts. ARPE treats this propagation as downstream plasticity rather than as a separate mechanism. The primary update occurs inside a bounded window, while related changes emerge as the updated structure participates in later cycles of activation.

2.6 Cascade Prevention

Cascade prevention describes the system's protective response when destabilization occurs without successful integration. If affective charge is too low, the structure may never open and no update occurs. If affective charge is high but new input is not coherently tagged, the structure may become unstable without becoming adaptive. If resonance error overwhelms the structure's coherence, the system may close the window, narrow the active structure, suppress integration, or route the material into compensatory stabilization.

This protective logic is essential because plasticity is not automatically beneficial. A system capable of rewriting itself must also prevent maladaptive rewriting, uncontrolled identity disruption, or the spread of instability across unrelated structures. ARPE

therefore includes cascade prevention as part of the core mechanism rather than as an optional clinical concern. The same process that permits change must also regulate when change is withheld, slowed, or contained.

Taken together, Sections 2.1 through 2.6 define ARPE as a threshold-governed plasticity mechanism. The model requires an identifiable target structure, a joint affect-error threshold, a bounded labile interval, affective tagging of new input, rewriting of internal weighting, restabilization, and containment of failed or excessive destabilization. These requirements distinguish ARPE from generic emotional arousal, ordinary learning, simple repetition, and short-term insight.

3.0 Internal Processing Dynamics

ARPE unfolds through a functional sequence rather than a rigid clocked process: a stabilized structure is destabilized, new input is affectively tagged, the structure is rewritten through weight propagation, and the updated form is restabilized through coherence lock. Depending on the domain, this sequence may occur over seconds, minutes, days, or repeated cycles of reactivation.

The processing dynamics are exploratory because the public manuscript does not claim a single fixed anatomical clock or universal parameter set. Instead, ARPE defines the structural logic that any empirical or computational rendering must preserve. A valid rendering must show how reactivation, affective charge, resonance error, input binding, and coherence restoration interact across a bounded window. It must also distinguish adaptive update from non-update, overload, fragmentation, and defensive closure.

3.1 Destabilization Trigger

The destabilization trigger occurs when resonance error and affective charge jointly reach sufficient magnitude to loosen the target structure from its stabilized state. The system does not destabilize every activated structure because doing so would make ordinary thought unstable. Instead, ARPE predicts selective destabilization only when the active structure is both meaningfully mismatched and affectively marked as important.

This trigger can arise through multiple routes. Corrective feedback may contradict an existing belief. A new relational experience may contradict an old attachment expectation. A successful action may contradict a long-standing self-limiting narrative. A traumatic memory may be reactivated in a safer context where the original threat mapping no longer fits. A skill error may reveal that a prior motor or cognitive strategy is no longer adequate. In each case, the relevant structure encounters a mismatch that is not merely intellectual but affectively consequential.

The destabilization phase has two interpretive boundaries. If affective charge is too low, the discrepancy may be noticed but treated as irrelevant, producing no structural opening. If affective charge is too high relative to coherence, the system may protect itself by narrowing attention, blocking integration, escalating avoidance, or fragmenting the experience. Adaptive ARPE processing therefore depends on a middle range in which mismatch is strong enough to matter but not so overwhelming that integration becomes impossible.

3.2 Tagging & Integration Window

Once the structure is labile, new input must be bound to it through affective tagging. This phase is the functional core of ARPE because it determines whether destabilization becomes adaptive plasticity or remains an unstable episode. The system must attach new meaning, valence, expectation, or relevance to the active structure while it is temporarily available for change.

The integration window is constrained by both availability and coherence. Availability means that the structure has become flexible enough for new input to matter; coherence means that the structure has not dissolved into disconnected affective fragments. ARPE predicts that adaptive change is most likely when the system can hold the old structure and the new input simultaneously, allowing comparison, retagging, and integration to occur before the window closes.

During this phase, input does not function as neutral information. It becomes meaningful because it is attached to the active structure under emotional salience. A corrective experience that occurs outside the active window may be remembered as an event but fail to rewrite the target structure. The same experience, encountered while the structure is labile, can become a new affective marker that changes how the structure is retrieved and interpreted later.

3.3 Rewriting & Weight Propagation

Rewriting begins when tagged input alters the internal weighting of the labile structure. The update may affect emotional valence, prediction strength, retrieval priority, behavioral readiness, or narrative role. ARPE treats these changes as weight propagation because the target structure is usually connected to adjacent structures. A modified memory can alter a belief; a modified belief can alter an intention; a modified intention can alter future behavior; and repeated behavioral confirmation can feed back into the narrative layer.

Weight propagation does not imply that every related structure changes at once. ARPE predicts graded propagation. Closely linked structures update first, while distant structures may shift only after repeated activation cycles. For example, a single corrective experience may soften the threat weighting of one memory, but a broader identity narrative may require multiple aligned updates before it reorganizes. This allows ARPE to explain both sudden breakthroughs and gradual developmental change within the same mechanism.

3.4 Restabilization & Coherence Lock

Restabilization occurs when the updated structure regains coherence and exits the labile interval. Coherence lock does not mean that the structure becomes permanently fixed. It means that the revised structure has become stable enough to function as the new standing form for future activation. After coherence lock, the system no longer treats the update as an open contradiction; it carries the revised structure forward as the current baseline.

Coherence lock can be adaptive, partial, or defensive. Adaptive lock occurs when the structure incorporates new input while preserving continuity. Partial lock occurs when some elements update while others remain tied to the prior structure. Defensive lock occurs when the system closes the window to prevent overload, leaving the old structure largely intact or creating a compensatory structure that restores short-term stability. These distinctions allow ARPE to model successful change, incomplete change, and failed change without treating them as unrelated outcomes.

Once coherence lock occurs, the updated structure becomes available to influence later perception, memory retrieval, action selection, and self-interpretation. This is where ARPE connects short-window reconsolidation to long-term psychological development. A brief window may alter one structure, but repeated coherence-locked updates can change the larger architecture through accumulation, reinforcement, and alignment across related structures.

3.5 Window Timing and Duration

ARPE does not assign one universal duration to the reconsolidation window. The window may be brief when the target structure is small, weakly embedded, or already near update readiness. It may extend across repeated reactivations when the structure is distributed across memories, beliefs, intentions, and identity narratives. In public form, the important commitment is not exact duration but boundedness: the structure enters a temporary state of increased plasticity and then returns to stability.

Window duration is expected to depend on affective charge, error magnitude, structure coherence, complexity of the structure, and availability of integration input. A simple memory correction may stabilize quickly. A self-narrative revision may require multiple smaller windows that align over time. A traumatic structure may open only partially, repeatedly closing when overload exceeds integration capacity. ARPE therefore treats timing as structure-dependent rather than as fixed across all domains.

For empirical use, window timing should be inferred from observable transition markers: emotional salience, mismatch awareness, temporary interpretive flexibility, responsiveness to new input, and later change in retrieval or behavior. The model does not require every study to observe the window directly, but it does require evidence that the target structure was reactivated, destabilized, retagged, and restabilized rather than merely discussed or rehearsed.

3.6 Partial Update States

Not all ARPE windows produce complete rewriting. Partial update occurs when some aspects of a structure accept new input while other aspects remain organized by the prior weighting. This may occur when affective charge is sufficient to open the window but integration input is narrow, when the structure is highly distributed, or when the system protects central identity assumptions while allowing peripheral details to change.

Partial updates are important because they explain mixed outcomes. A person may feel less emotional distress but still behave according to an old belief. A memory may become less vivid while the associated avoidance pattern remains. An intention may feel more possible but not yet become automatic. A narrative may soften without fully reorganizing. ARPE treats these as intermediate states rather than failures of the model.

The model predicts that partial updates should become visible as uneven change across measures. Emotional reactivity may shift before behavior changes. Verbal belief endorsement may change before embodied response changes. Behavioral flexibility may appear in safe contexts before appearing under stress. This staged pattern does not contradict ARPE; it suggests that different components of the structure are locking at different rates.

3.7 Failed Update and Defensive Closure

Failed update occurs when the structure becomes destabilized but does not incorporate new input into a coherent revision. This can happen when affective charge overwhelms coherence, when input is ambiguous, when the structure is too central to identity to revise quickly, or when competing structures block integration. In such cases, the system may close the window without change or strengthen protective responses that prevent future destabilization.

Defensive closure is not treated as irrational within ARPE. It is a stability-preserving response. If the system estimates that rewriting would create more incoherence than adaptation, it may preserve the old structure even when contradiction is present. This accounts for cases where people recognize evidence against a belief, recall a painful memory in a safer context, or experience insight, yet still return to the previous structure afterward.

The model predicts that failed updates will leave traces: increased avoidance, heightened sensitivity to future reactivation, sharper defensive narratives, or repeated cycles of near-change followed by collapse. These traces are not random byproducts. They indicate that the system approached plasticity but did not complete integration. In later empirical work, such traces may help distinguish non-opening from failed opening.

3.8 Cascade Containment

Cascade containment limits the spread of destabilization across related structures. Because cognitive-emotional structures are interconnected, rewriting one structure can place pressure on others. A revised memory may challenge a belief; a revised belief may destabilize an identity role; a revised role may alter narrative continuity. ARPE predicts that the system regulates this spread by allowing bounded propagation while preventing uncontrolled collapse.

Containment may occur through narrowing the active structure, delaying downstream integration, preserving parts of the old structure temporarily, or routing unresolved material into later windows. This allows large psychological change to occur without requiring all linked structures to update simultaneously. In developmental and therapeutic contexts, this may explain why durable change often appears as a sequence of linked but bounded shifts rather than as one total transformation.

A containment failure would appear as spreading instability: related memories, beliefs, intentions, and narratives become destabilized faster than they can be restabilized. ARPE treats this as a failure mode rather than as successful plasticity. Adaptive change is therefore not defined by the amount of destabilization produced, but by whether destabilization remains bounded enough to permit coherent restabilization.

3.9 Long-Scale Propagation

ARPE links short-window plasticity to long-scale psychological development through propagation across repeated cycles. A single window may produce a local update, but durable transformation occurs when multiple updated structures begin to align. Over time, a series of memory updates, belief revisions, intention shifts, and narrative retaggings can reorganize the system's broader pattern of prediction and response.

This long-scale process helps ARPE explain why some changes feel sudden while others unfold gradually. A sudden change may reflect the visible completion of a sequence whose earlier windows prepared the structure for final rewriting. Gradual change may reflect many small updates that accumulate without a single dramatic breakthrough. In both cases, the underlying mechanism remains the same: bounded destabilization, affective tagging, rewriting, and restabilization repeated across time.

The exploratory implication is that growth, recovery, learning, and identity transformation may be understood as patterned ARPE sequences rather than as separate categories of change. The specific domain differs, but the update logic remains consistent. ARPE therefore describes how stabilized structures become revisable and how revised structures become the basis for future perception, intention, interpretation, and self-organization.

4.0 Applications and Predictions

ARPE generates testable predictions across clinical, developmental, learning, and experimental domains because it treats durable change as a bounded structural update rather than as a vague improvement in mood, insight, or performance. The central empirical expectation is that lasting change should occur most reliably when a target structure is reactivated, affective charge rises, resonance error becomes salient, new input is integrated during the labile interval, and the updated structure restabilizes with measurable downstream effects.

The framework does not predict that emotional intensity alone produces adaptive change. It predicts that emotional intensity becomes plasticity-relevant only when it is coupled to an active structure and a meaningful error signal. Likewise, contradiction alone is not sufficient. A person may receive corrective information many times without reorganization if the information fails to contact the target structure at the affective level. ARPE therefore makes a discriminating prediction: interventions or experiences that combine reactivation, affective salience, and structured corrective input should outperform interventions that provide only one or two of those elements.

A minimal falsifiable hypothesis follows from this mechanism: when a target structure is reactivated under affective salience and meaningful mismatch, then paired with coherent corrective input, delayed measures should show greater structure-specific

change than reactivation alone, affective induction alone, or corrective information alone. If those partial conditions perform equivalently across repeated controlled tests, ARPE's distinct mechanism is weakened.

4.1 Structured Reconsolidation Timing

Prediction: Targeted memory reactivation paired with emotionally meaningful corrective input should produce more durable increases in autobiographical coherence, intention stability, and affective flexibility than reactivation alone or positive emotion alone. In ARPE terms, the effect should be strongest when the old structure is active, mismatch becomes affectively salient, new input is bound during the labile interval, and the updated structure restabilizes before defensive closure occurs.

A mechanism-level prediction follows from this structure: the strongest change should occur when the timing of corrective input coincides with the open window rather than before or after it. Corrective reassurance offered before the target memory or belief is active may fail to bind to the structure. Corrective input offered after defensive closure may be understood intellectually but not incorporated structurally. The model therefore predicts an interaction between reactivation timing, affective intensity, and the presence of coherent replacement input.

Expected measures include changes in autobiographical coherence ratings, reduced contradiction between stated intention and later behavior, lowered physiological reactivity when the original material is reactivated, and reduced recurrence of the prior affective interpretation. A falsifying pattern would occur if reactivation plus affectively meaningful corrective input performs no better than neutral discussion, simple reassurance, or exposure without retagging across repeated controlled comparisons.

4.2 High-Affect Memory Updating

Prediction: Reactivation of emotionally charged traumatic memories inside a controlled ARPE window should reduce intrusive elements, physiological reactivity, and spontaneous recall frequency more reliably than reactivation without coherent retagging. The model does not predict that intense traumatic activation alone is therapeutic. It predicts that traumatic activation becomes plasticity-relevant only when the structure is held in a bounded labile state and paired with input that updates its threat weighting without overwhelming coherence.

In trauma-related applications, ARPE distinguishes between exposure, flooding, reconsolidation, and integration. Exposure may reduce response through habituation, but ARPE specifically predicts structural change when the memory's affective tag and predictive meaning are rewritten. Flooding may destabilize without integration, increasing defensive closure or fragmentation. Integration occurs when the memory remains identifiable while receiving new emotional and interpretive weighting inside the window.

Observable outcomes should include lower startle or autonomic response to reactivation cues, decreased intrusive recurrence, improved narrative sequencing, and increased ability to recall the event without recreating the original affective state. A falsifier would be repeated evidence that intense reactivation plus coherent corrective input produces no greater structural change than exposure alone, or that change occurs without any detectable reactivation, affective salience, or retagging sequence.

4.3 Skill Acquisition & Belief Updating

Prediction: Deliberate practice that induces manageable prediction error plus positive or mastery-linked affect will accelerate schema updating and produce steeper learning curves with stronger long-term retention than neutral repetition. In ARPE terms, skill acquisition becomes more durable when the learner's existing motor, perceptual, or conceptual structure is destabilized by error and then retagged with successful correction before restabilization.

This prediction explains why repetition without feedback can plateau. If practice repeats the same stable structure without sufficient error, no update window opens. If error is too high, the learner may lose coherence, disengage, or defensively simplify the task. The strongest learning should occur inside an optimal error-affect zone: enough mismatch to destabilize the old strategy, enough affective reward or relevance to mark the correction, and enough coherence to lock the new pattern.

For belief updating, ARPE predicts that contradiction changes beliefs most reliably when it becomes personally, emotionally, or identity-relevant. Neutral counterevidence may be dismissed as abstract. Emotionally threatening counterevidence may trigger defensive closure. But counterevidence that is affectively salient while still integrable should weaken prior certainty and allow belief retagging. Measures may include reduced confidence rigidity, improved transfer to new contexts, and altered behavior under conditions that previously activated the old belief.

4.4 Longitudinal Psychological Growth

Prediction: Individuals engaging in regular reflective practices that reliably produce controlled ARPE windows will show progressive increases in coherence scores, behavioral flexibility, and narrative integration across 6–12 months. The mechanism is not reflection alone. The relevant condition is repeated reactivation of meaningful structures under manageable affective charge, followed by reinterpretation, retagging, and restabilization.

Longitudinal growth under ARPE should appear as cumulative alignment across structures. Early windows may target local memories or beliefs; later windows may reorganize intentions, roles, and self-narratives. A person may first reinterpret an event, then act differently in a related context, then update the belief attached to that action, and eventually revise the narrative that connects those experiences. Growth is therefore modeled as chained structural updating rather than as a global trait increase.

Expected measures include higher temporal coherence in life-story interviews, reduced contradiction between self-report and behavior, broader context transfer of new beliefs or intentions, and greater recovery after emotional perturbation. A falsifying pattern would occur if reflective practice improves mood or verbal insight without producing any durable structural change in memory weighting, belief flexibility, intention stability, or behavior across time.

These predictions are directly evaluable with reactivation plus emotion protocols, psychophysiological measures, behavioral transfer tests, structured self-report, delayed recall tasks, and narrative coherence analysis. Neural measures such as EEG or fMRI may be useful in later validation stages, but ARPE does not depend on any single measurement channel.

For every prediction below, the appropriate comparison is not simply change versus no change. The stronger comparison is complete ARPE sequencing versus partial sequencing: reactivation alone, affective induction alone, corrective information alone, and combined reactivation-affect-error-tagging conditions. This keeps the framework testable by showing which element is doing the explanatory work.

4.5 Prediction-to-Measure Mapping

ARPE condition	Expected signature	Candidate measures
Threshold opening	Affective salience and mismatch rise together.	Self-report intensity, autonomic arousal, error awareness, and neural markers where independently specified.
Affective tagging	New input becomes bound to the active structure.	Post-reactivation appraisal shift, recall content change, valence rating change.
Restabilization and delayed persistence	Updated structure persists after the window closes and transfers beyond the immediate session.	Delayed recall, behavior transfer, reduced reactivity, coherence ratings.

This mapping is intentionally broad. ARPE does not require one privileged measurement channel. Different studies may operationalize the same sequence using behavioral, physiological, neural, or narrative measures, provided the design preserves the core temporal relation among reactivation, affective charge, mismatch, tagging, rewriting, and restabilization.

4.6 Falsification Conditions

ARPE is weakened if durable structural change occurs without any detectable reactivation of the target structure, without affective salience, without mismatch or error, and without any evidence of retagging or restabilization. It is also weakened if complete ARPE conditions fail to outperform partial conditions across repeated controlled studies.

A strong falsification test should isolate the mechanism's required elements. If reactivation without affective charge produces the same durable change as complete ARPE conditions, the affective-gating claim is weakened. If affective induction without target-structure reactivation produces the same change, the structure-specific claim is weakened. If contradiction without retagging produces the same outcome, the tagging requirement is weakened. If immediate change appears but does not persist after restabilization, the model should treat the result as temporary activation rather than structural update.

ARPE is also constrained by negative cases. If a target structure is repeatedly reactivated under conditions that appear to meet the threshold but remains unchanged across behavioral, affective, and narrative measures, the framework must lower its claim: either the threshold was misidentified, the integration input failed to bind, or the update assumptions are incomplete. The model should not treat every non-change as hidden change. Observable non-update is an important boundary condition.

ARPE does not claim that every durable change is caused by affect-gated reconsolidation. It claims that when durable structure-specific change follows reactivation, affective salience, meaningful mismatch, retagging, and delayed restabilization, ARPE provides a candidate mechanism worth testing against simpler alternatives.

4.7 Minimal Experimental Protocol

A minimal ARPE protocol should begin by identifying a target structure before intervention and establishing a baseline for how it activates, what affect accompanies it, what interpretation or behavior it produces, and how stable it remains across repeated exposures.

The next phase should manipulate the elements of the ARPE sequence separately. One condition should reactivate the target structure without adding affective charge. A second should induce affective charge without reactivating the target structure. A third should present corrective input without creating a strong mismatch. A fourth should combine reactivation, affective charge, mismatch, and corrective input in one controlled sequence. ARPE predicts that the fourth condition should produce the strongest delayed and structure-specific update.

The protocol should include immediate and delayed measurements. Immediate measures establish whether the structure was activated and whether affective salience was present. Delayed measures determine whether an actual structural update occurred. A change that appears only immediately after the session should be treated as activation, mood shift, compliance, or temporary appraisal unless it persists after restabilization. A change that persists across delay and transfers to related contexts is more consistent with ARPE.

A minimal protocol should therefore preserve the following sequence: target identification, baseline activation, controlled destabilization, input tagging, delayed retest, transfer test, and failure classification. This sequence allows studies to distinguish adaptive update from non-opening, partial update, defensive closure, and nonspecific improvement. The core endpoint is delayed structure-specific change, not immediate emotional intensity or short-term report of insight.

4.8 Competing Explanations and Discriminator Logic

ARPE must be separated from ordinary learning, habituation, persuasion, compliance, mood alteration, and general arousal. None of these outcomes necessarily require a labile reconsolidation window or affective retagging of a target structure.

The discriminator is delayed, structure-specific change. Habituation predicts reduced response with repeated exposure, persuasion predicts altered verbal belief statements, and mood enhancement predicts temporary improvement. ARPE predicts that the old structure activates differently after restabilization because its affective tag, expectation, or internal weighting has changed.

A strong study should therefore compare ARPE against partial explanations rather than against no treatment alone. It should include a reactivation-only condition, an affect-only condition, an information-only condition, and a combined condition. If all conditions produce equivalent outcomes, the distinctiveness of ARPE is weakened. If the combined condition uniquely produces

delayed structure-specific updating, ARPE gains support as a mechanism of bounded plasticity rather than as a label for generic improvement.

4.9 Exploratory Use Cases

In clinical contexts, ARPE may help organize why some interventions produce durable change while others produce temporary relief. The model would predict that interventions are more likely to produce structural change when they reactivate a target structure, generate manageable affective error, provide corrective input, and support restabilization. This does not imply that ARPE is itself a treatment protocol. Rather, it provides a mechanism-level lens for evaluating when and why different interventions may create actual plasticity.

In learning and performance domains, ARPE suggests that optimal practice is not merely repetition but structured destabilization followed by successful retagging. A learner must encounter error that matters, receive input that resolves the error, and restabilize around the updated pattern. This helps explain why challenging but manageable practice can produce durable skill growth while either effortless repetition or overwhelming failure produces weaker update.

In identity development, ARPE provides a way to understand transformation as a sequence of bounded updates rather than a sudden replacement of the self. A person may update one memory, then one belief, then one intention, then one role narrative. Over time, these linked restabilizations can produce a larger identity shift while preserving continuity. This makes ARPE useful for describing gradual growth, recovery, and maturation without reducing them to generic self-improvement.

5.0 Limitations and Future Work

ARPE is currently a functional and exploratory mechanism model rather than a fully parameterized operational system. The public manuscript defines the structural logic of emotion-gated reconsolidation without fixing timing constants, baseline plasticity rates, neural implementation details, or measurement thresholds. Empirical versions must therefore determine how to identify window opening, tagging success, restabilization, partial update, and defensive closure in specific study designs.

A second limitation concerns boundary identification. The framework assumes that cognitive-emotional structures can be meaningfully identified across reactivation, destabilization, rewriting, and restabilization. In practice, structures may be distributed, overlapping, or difficult to separate from adjacent memories, beliefs, intentions, and narratives. A study that cannot identify the target structure with sufficient clarity cannot determine whether ARPE has occurred or whether observed change reflects mood variation, demand effects, generic learning, or unrelated contextual influence.

A third limitation is that affective charge is not automatically beneficial. Strong emotion may open update potential, but it may also trigger avoidance, defensive closure, dissociation, rumination, or cascade instability. ARPE therefore should not be interpreted as a simple intensification model. The central question is not whether emotion is high, but whether emotion is structured, bounded, and paired with integrative input while the target structure remains coherent enough to update.

5.1 Boundary Conditions

ARPE applies only where an identifiable cognitive-emotional structure can be reactivated and where that structure can plausibly become labile under affective error. It does not apply to every emotional reaction, every memory recall, every learning event, or every therapeutic improvement. Events outside ARPE include generic arousal without target-structure activation, ordinary repetition without mismatch, intellectual insight without retagging, and temporary mood change without durable restabilization.

The model also does not claim that every durable change requires a single dramatic emotional episode. Some ARPE sequences may be distributed across repeated smaller windows. However, even gradual change must preserve the basic pattern: reactivation, affective error, tagging, rewriting, and restabilization. If gradual change occurs without any identifiable structure-specific update sequence, it should be explained through broader learning, habituation, social reinforcement, or developmental mechanisms rather than through ARPE alone.

5.2 Measurement Risks

The strongest measurement risk is mistaking short-term activation for structural change. A person may report insight, relief, distress, certainty, motivation, or clarity immediately after an emotionally salient event, yet return to the prior structure later. ARPE therefore requires delayed measurement. A valid study should look for persistence after the presumed window closes, transfer to related contexts, reduced recurrence of the old structure, or altered behavior under conditions that previously activated the old pattern.

Another risk is relying on one measurement channel. Self-report may capture conscious appraisal but miss defensive behavior. Physiology may show arousal reduction without narrative integration. Neural measures may reveal state change without identifying the target structure. Behavioral measures may show transfer without clarifying whether retagging occurred. ARPE is best tested through convergent evidence across at least two domains, such as report plus behavior, physiology plus recall, or narrative coherence plus longitudinal transfer.

5.3 Competing Explanations

ARPE must be distinguished from simpler explanations by the same discriminator used throughout the manuscript: durable, structure-specific update. If change remains limited to mood, verbal endorsement, compliance, or short-term performance, competing explanations remain more parsimonious.

If the old structure no longer activates in the same way, related behavior changes under comparable conditions, and the new structure persists beyond the immediate emotional state, ARPE becomes a stronger explanation.

A PI-level test of the framework should therefore prioritize discriminating designs over confirmatory language. The strongest early studies would not attempt to prove the entire ARPE model at once; they would test whether the proposed sequence explains delayed structure-specific change better than simpler alternatives under controlled comparison conditions.

5.4 Future Development Roadmap

Future development should proceed in layers. The first layer should define reliable indicators of target-structure activation and window opening. The second layer should test whether affective charge and resonance error interact as predicted. The third layer should compare complete ARPE conditions against partial conditions. The fourth layer should track delayed restabilization and transfer. The fifth layer should model individual differences in plasticity rate, defensive closure, and cascade sensitivity.

Computational development should treat the current pseudo-equation as a structural starting point, not as a finished algorithm. Simulations may vary threshold shape, affect-error interaction, coherence decay, retagging strength, and propagation depth. The goal of simulation would not be to replace empirical study, but to generate distinguishable update regimes: non-opening, adaptive update, partial update, defensive closure, cascade instability, and long-scale reorganization.

Future directions include:

- Multimodal reactivation studies to map candidate ARPE window signatures across self-report, behavior, physiology, and optional neural measures.
- Computational instantiations of the ARPE Update Structure defined in Section 2.2.
- Controlled studies testing ARPE-guided hypotheses for trauma-related memories, depression-linked beliefs, and maladaptive self-models, with safety and containment criteria defined in advance.
- Developmental extensions tracking ARPE maturation from adolescence through late life.

5.5 Operationalization Gaps

A major open problem for ARPE is operationalizing the difference between activation, opening, tagging, rewriting, and restabilization. These stages are structurally distinct in the model, but empirical work must determine which observable markers

reliably separate them. Without this separation, a study may mistake recall for destabilization, emotional intensity for tagging, behavioral compliance for rewriting, or short-term relief for restabilization.

For this reason, future operational versions should use stage-specific indicators. Activation may be indexed by recall content, cue responsiveness, or explicit recognition. Opening may be indexed by increased flexibility, contradiction salience, or temporary instability of prior interpretation. Tagging may be indexed by valence shift or new appraisal binding. Rewriting may be indexed by altered retrieval, expectation, or behavior. Restabilization may be indexed by persistence after delay and transfer across contexts.

5.6 Individual Differences and Plasticity Profiles

ARPE also requires a more detailed treatment of individual differences. People may vary in window-opening sensitivity, affective tolerance, coherence preservation, defensive closure tendency, and propagation depth. One person may update rapidly from a single emotionally meaningful contradiction, while another may require repeated smaller windows before the same structure becomes revisable. These differences should not be treated as noise; they may be central parameters in later ARPE models.

Possible plasticity profiles include low-opening profiles, where structures resist destabilization; high-opening profiles, where structures become labile easily; high-containment profiles, where update is permitted but spread is tightly controlled; and cascade-sensitive profiles, where one update rapidly destabilizes adjacent structures. Mapping these profiles would help explain why similar interventions produce different outcomes across individuals.

5.7 Ethical and Clinical Caution

Because ARPE concerns emotion-driven structural change, it carries ethical and clinical risks if misapplied. Opening a reconsolidation window without adequate containment may increase distress, strengthen defensive closure, or destabilize related structures. The framework should therefore not be read as a recommendation to intensify emotional material indiscriminately. Any clinical translation must prioritize safety, consent, pacing, stabilization, and the distinction between controlled reactivation and overwhelming exposure.

The ethical risk is especially important for trauma, identity disruption, and high-affect belief systems. In those domains, destabilization may outpace integration. ARPE therefore requires future protocols to include stopping rules, distress monitoring, delayed follow-up, and clear criteria for distinguishing adaptive plasticity from destabilizing reactivation. The model's value depends on preserving the boundary between mechanistic explanation and unsafe intervention.

5.8 Implementation Limits

The public ARPE manuscript does not define runnable algorithms, parameter defaults, or intervention procedures. Its pseudo-formal expression is intended to preserve structural clarity, not to serve as executable code. Any implementation would require additional decisions about how to estimate affective charge, how to quantify resonance error, how to define the threshold, how to measure coherence, and how to determine whether restabilization has occurred.

Implementation should also avoid assuming that the same parameter values apply across memory, belief, skill, intention, and narrative domains. A motor-learning structure, an autobiographical memory, and an identity narrative may all follow ARPE logic while requiring different time windows, measurement channels, and update thresholds. The framework is unified at the mechanism level, not at the level of one universal implementation recipe.

5.9 Staged Validation Priorities

Validation should begin with narrow, controlled cases before moving to broad psychological transformation claims. The first priority is to demonstrate that complete ARPE conditions produce more durable structure-specific change than partial controls. The second priority is to identify which measures best distinguish opening, tagging, rewriting, and restabilization. The third priority is to test whether linked structures update in the graded propagation pattern predicted by the model.

Only after these narrow tests should ARPE be applied to large-scale constructs such as identity transformation, long-term recovery, or lifespan development. Those domains are likely to contain many overlapping mechanisms, and ARPE should earn its role by explaining measurable update sequences rather than by absorbing every form of change into its vocabulary. This staged validation protects the framework from becoming too broad to falsify.

6.0 Final Synthesis

The Affective Reconsolidation and Plasticity Engine offers an exploratory account of how cognitive-emotional structures may become revisable, update, and restabilize across time. By centering long-term change on bounded reconsolidation windows and the ARPE Update Structure, the framework explains how memories, beliefs, intentions, schemas, and personal narratives may be rewritten while preserving enough coherence for continuity.

Its central contribution is to separate change from activation. A structure may be activated without opening, opened without integrating, or partially updated without reorganizing the larger system. By distinguishing these outcomes, ARPE explains why some experiences produce durable update while others produce temporary feeling, repeated rumination, or defensive return to the prior pattern.

The inclusion of a pseudo-formal update structure, clear destabilization–tagging–restabilization phases, and directly testable predictions positions ARPE as an independent framework for emotion-driven structural plasticity. While further refinement of window timing, measurement thresholds, and individual calibration remains necessary, ARPE provides a practical and falsifiable lens for studying durable psychological updating.

As an exploratory framework, ARPE should be read as a disciplined starting point rather than a completed empirical proof. Its value lies in making the conditions of structural change explicit: the target structure must be active, the mismatch must matter, the affective charge must open a bounded window, new input must be tagged into that window, and the revised structure must restabilize in a form that can be observed later. Every future test of ARPE should preserve that sequence or state clearly which element is being challenged.

Appendix A: Symbol Definitions

The following symbols appear in the ARPE Update Structure (Section 2.0) and throughout the processing phases described in Section 3.0.

Symbol	Operational definition
S(t)	The current cognitive-emotional structure — a memory trace, belief, intention, schema, or narrative element held in its stabilized form.
I(t)	New information available while the structure is labile, including sensory evidence, corrective feedback, or reinterpretation bound to the structure during tagging.
A(t)	The emotional intensity attached to the current reactivation, functioning as the condition under which a structure is destabilized and retagged rather than left intact.

E(t)	Resonance error — the mismatch between the current structure and incoming input that marks the structure as a candidate for revision.
C(t)	The current coherence of the target structure. As coherence strengthens, the window closes and the structure resists further update; as it weakens, more of the structure remains available for rewriting.
θR	The reconsolidation threshold. Rewriting occurs only when affective charge and resonance error jointly meet or exceed this boundary.
S'(t+1)	The updated structure carried forward once restabilization completes, serving as the standing form against which subsequent reactivations are compared.

Appendix B: Structural Constraints

Element	Conceptual requirement	Interpretive description
Reconsolidation window	Updating occurs only within a bounded labile interval.	Once the window closes, the structure resists change until it is destabilized again.
Destabilization threshold	Affective charge and resonance error must jointly reach sufficient magnitude.	Below this point the structure remains stable and no rewriting occurs, however often it is reactivated.
Affective tagging	New input must be bound to the structure under an updated affective marker.	Where tagging does not occur, the structure is destabilized without being coherently rewritten.
Cascade prevention	Failed or partial updates must be able to trigger a protective response.	Where integration does not complete, compensatory loops restore stability rather than allowing fragmentation to propagate.

The framework remains coherent only if its defining conditions are preserved: bounded reconsolidation windows, affect-gated destabilization, identifiable structure content, tagging-based rewriting, and completed restabilization. Where these conditions fall away, the resulting account no longer preserves ARPE as described here.

Appendix C: Update Regimes

Regime	Defining condition	Interpretive meaning
Non-opening	Reactivation occurs below threshold.	The structure remains stable and no rewriting occurs.
Adaptive update	Threshold is met and input is coherently tagged.	The structure is rewritten and restabilizes in updated form.

Partial update	Some components update while others retain prior weighting.	Change appears uneven across emotion, belief, behavior, and narrative.
Defensive closure	The system closes the window to preserve coherence.	The old structure remains protected despite contradiction.

Appendix D: Exploratory Measurement Architecture

Appendix D defines a measurement architecture for exploratory ARPE studies. It separates the construct into five observable stages: target-structure activation, threshold opening, affective tagging, rewriting, and restabilization. Each stage should be measured with at least one primary indicator and one secondary indicator so that studies do not rely on a single channel. Activation may be indexed by recall content and cue responsiveness; opening by contradiction salience and affective intensity; tagging by appraisal shift and valence change; rewriting by altered retrieval, expectation, or behavior; and restabilization by delayed persistence and transfer. A study should report which stages were observed, which were inferred, and which were not supported.

Appendix E: Plasticity Profiles

Plasticity profile	Defining pattern	Expected risk	Research use
Low-opening profile	Structures resist destabilization even when mismatch is present.	Non-opening or minimal update despite repeated activation.	Useful for studying resistance to structural change and threshold sensitivity.
High-opening profile	Structures enter labile states easily under affective mismatch.	Over-opening, instability, or premature destabilization.	Useful for studying affective sensitivity and window-control requirements.
High-containment profile	Local update is permitted while spread to adjacent structures is limited.	Partial update or slow propagation across related structures.	Useful for studying staged change, safety, and bounded propagation.
Cascade-sensitive profile	One update rapidly destabilizes linked structures.	Cascade instability or difficulty restabilizing the broader system.	Useful for studying containment, failure modes, and propagation limits.

Appendix F: Minimal Reporting Checklist

Reporting item	Minimum requirement
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Target structure	Define the memory, belief, intention, schema, behavior pattern, or narrative element being tested.
Baseline activation	Measure how the target structure activates before any update condition is introduced.
Affective charge	Specify how affective intensity is observed, measured, or inferred.
Mismatch or error	Define the meaningful mismatch between the active structure and incoming evidence, context, outcome, or interpretation.
Tagging condition	Describe the corrective input, reinterpretation, or experience intended to bind to the labile structure.
Delayed restabilization	Test whether change persists after the immediate session or activation period has ended.
Partial-condition controls	Include controls for reactivation alone, affective induction alone, corrective information alone, and the combined ARPE sequence when feasible.
Non-update and failure cases	Report non-opening, partial update, defensive closure, and failed update outcomes rather than omitting them.